Andrew MM Matheson

New York, NY | am5835@columbia.edu | AMMmatheson | andrewmatheson.bsky.social

Summary

Neuroethologist studying environmental plasticity in the Iberian ribbed newt.

Education

PhD **New York University**, Neuroscience and Physiology — New York, NY, USA
- Doctoral research advised by Katherine I Nagel — Jan 2015 – Jan 2021

BSc **McGill University**, Computer Science and Biology — Montreal, QC, Canada
- Research on vocal communication in Jon T Sakata's lab — Jan 2011 – Jan 2015

Experience

Tosches Lab, Department of Biological Sciences, Columbia University, Postdoctoral Research Scientist — New York, NY
Jan 2021 – present
5 years 6 months
- Investigating neural mechanisms of environmental plasticity and sensory processing in the Iberian ribbed newt.
- Neural circuit plasticity research
- Electrophysiology techniques
- Computational neuroscience

Publications

Iberian ribbed newts — Jan 2025
Matheson AMM, Chua NJ, Tosches MA
10.1016/j.cub.2024.11.062 (Current Biology)

A neural circuit for wind-guided olfactory navigation — Jan 2022
Matheson AMM, Lanz AJ, Currier TA, Licata AM, Syed MH, Nagel KI
10.1038/s41467-022-32247-7 (Nature Communications)

A population of central complex neurons promoting airflow-guided orientation in Drosophila — Jan 2020
Currier TA, Matheson AMM, Nagel KI
10.7554/eLife.61510 (eLife)

Central neurons encoding wind direction in Drosophila — Jan 2019
Suver MP, Matheson AMM, Sarkar S, Damiata M, Schoppik D, Nagel KI
10.1016/j.neuron.2019.03.012 (Neuron)

Cortical control of vocal interaction in neotropical singing mice — Jan 2019
Okobi DE, *Banerjee A*, Matheson AMM, Phelps SM, Long MA
10.1126/science.aau9480 (Science)

Relationship between the sequencing and timing of vocal motor elements in birdsong — Jan 2015
Matheson AMM, Sakata JT
10.1371/journal.pone.0143203 (Plos One)

Awards

Doupe Fellowship — Jan 2026
Awarded by the McKnight Endowment Fund for Neuroscience.

Young Investigator Award Jan 2026

Awarded by the International Society for Neuroethology. Award recognizing early-career achievements in neuroethology.